

Heart Beat and SIS-10 Framework: Drift Adaptive, Byzantine Resilient, Graph-Probabilistic, Control Observable System

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Abstract—The final integrated HB (Heart Beat) SIS-10 framework (2026) is introduced as a regulator grade heartbeat and safety kernel architecture for safety critical distributed systems that is distributionally robust, drift adaptive, multiscale, Byzantine resilient, graph probabilistic, and control observable. The framework replaces brittle deterministic invariants with stochastic and distributionally robust guarantees, thereby unifying network statistics, spectral graph connectivity, probabilistic inference, and control theoretic observability into a single coherent and formally verifiable safety structure.

The architecture further incorporates explicit parameter optimization, adaptive multiscale temporal reasoning, and formal verification hooks suitable for assurance critical deployment. The analysis demonstrates that Wasserstein ambiguity sets, spectral graph structure, adaptive temporal windows, and observability constraints together constitute an irreducible safety foundation for resilient SIS-10 governance under uncertainty, adversarial behavior, partial observability, and non stationary system drift.

In addition, the paper formally identifies the structural deficiencies of existing heartbeat and safety kernel designs including deterministic fragility, weak graph awareness, inadequate adversarial tolerance, poor drift adaptation, and limited observability and derives the principled architectural conditions required for next generation safety critical systems to maintain robustness, stability, and operational integrity under real world stochastic environments.

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These engagements provided direct exposure to real world operational safety systems and risk variability in safety critical environments.

I. INTRODUCTION

Safety critical distributed systems increasingly depend on heartbeat protocols and monitoring kernels to maintain availability, integrity, coordination, and fault containment under uncertainty, adversarial interference, and non-stationary operating conditions. However, existing architectures are predominantly based on deterministic invariants that degrade under stochastic drift, heavy tailed disturbances, partial observability, and Byzantine faults. These limitations reveal a structural mismatch between classical safety assumptions and the behavior of real world distributed systems.

This paper introduces the final integrated HB-SIS-10 framework (2026), a regulator grade heartbeat and safety kernel architecture designed to address this gap. The framework unifies distributionally robust statistics, drift adaptive modeling, multi scale temporal reasoning, graph probabilistic connectivity, and control theoretic observability into a single coherent and formally analyzable safety specification. Its central principle is the replacement of brittle deterministic invariants with stochastic, distributionally robust guarantees that remain valid under uncertainty and adversarial conditions.

The architecture is organized around six irreducible safety dimensions: (i) distributional robustness via Wasserstein ambiguity sets, (ii) drift adaptivity through bounded stochastic drift models, (iii) multi scale temporal stability via hierarchical window fusion, (iv) Byzantine resilience via robust aggregation and fjk connectivity constraints, (v) graph probabilistic resilience via spectral connectivity and reliability coupling, and (vi) control observability via rank constrained state reconstruction

conditions. Together, these dimensions define a unified safety envelope for distributed systems operating under uncertainty.

The framework formalizes the HB-SIS-10 model, constructs a stochastic heartbeat invariant, and defines a distributionally robust parameter optimization problem governing system adaptation under uncertainty. It further establishes a stochastic connectivity theorem linking spectral graph properties, reliability structure, and observability conditions, and develops a decomposition of total system safety risk into interpretable failure modes, including temporal instability, probabilistic disconnectivity, adversarial corruption, and observational incompleteness.

The analysis demonstrates that Wasserstein ambiguity structure, spectral graph connectivity, adaptive temporal fusion, and observability constraints jointly form an irreducible foundation for resilient SIS-10 governance. Any further improvement in robustness requires changes to fundamental modeling assumptions, including stochastic versus deterministic representations, bounded versus unbounded drift regimes, and the admissible class of adversarial perturbations.

DEFINITION 1 (HB-SIS-10 KERNEL)

Let $\mathcal{N} = \{1, \dots, N_{\max}\}$ be a finite set of nodes and $k \in \mathbb{Z}_+$ a discrete time index. The HB-SIS-10 kernel is defined as a coupled stochastic-drift dynamical system

$$\mathcal{K}_i(k) := (\mathcal{P}_i(k), D_i(k), H_i(k))_{i \in \mathcal{N}}$$

with the following structural axioms:

a) Axiom 1 (Distributional Uncertainty).: Each node i admits a latent distributional state $P_i(k)$ constrained by a Wasserstein ambiguity set:

$$\mathcal{P}_i(k) = \left\{ P : W_1(P, \hat{P}_i(k)) \leq \rho_i \right\}.$$

b) Axiom 2 (Bounded Drift Dynamics).: Each node evolves under a bounded drift process:

$$D_i(k+1) \subseteq \mathcal{T}(D_i(k)) \oplus \delta_i.$$

c) Axiom 3 (Coupled Evolution Law).: Distributional evolution is drift conditioned:

$$P_i(k+1) \in \Phi(\mathcal{P}_i(k), D_i(k)).$$

d) Axiom 4 (Heartbeat Functional).: Observed heartbeat signals are stochastic functionals of latent distributions:

$$H_i(k) = \mathcal{H}(P_i(k)) + \epsilon_i(k).$$

e) Axiom 5 (Worst Case Decision Semantics).: All safety evaluations are taken over admissible sets:

$$\inf_{P_i \in \mathcal{P}_i(k)} \sup_{D_i \in \mathcal{D}_i(k)} \mathbb{E}[\cdot].$$

Induced Stochastic Connectivity Structure

Define a time varying weighted graph

$$\mathcal{G}(k) = (\mathcal{N}, \mathcal{E}(k), W(k))$$

where edge weights are distribution-drift dependent:

$$w_{ij}(k) = \Psi(\mathcal{P}_i(k), \mathcal{P}_j(k), D_i(k), D_j(k)).$$

Let $\mathcal{L}(k)$ be the associated graph Laplacian and $\lambda_2(\mathcal{L}(k))$ its algebraic connectivity.

f) Stochastic Connectivity Theorem (SIS-10 Core Result).: Under Axioms 1–4, and bounded drift $\delta_i < \delta^*$, the system satisfies:

$$\mathbb{P}(\lambda_2(\mathcal{L}(k)) > 0) \geq 1 - \exp(-\alpha \cdot \rho_{\min} + \beta \cdot \|\delta\|),$$

for some constants $\alpha, \beta > 0$, where $\rho_{\min} = \min_i \rho_i$.

g) Interpretation.: Connectivity is not deterministic but distributionally persistent: graph disconnection becomes a tail event governed jointly by Wasserstein radius (uncertainty) and drift budget (non-stationarity).

Kernel Role

The HB-SIS-10 kernel is the minimal stochastic object generating:

- distributionally robust inference,
- drift aware evolution,
- probabilistic heartbeat consistency,
- and connectivity persistence guarantees.

All higher level SIS-10 results (spectral stability, observability, Byzantine resilience) are derived as corollaries of this kernel structure.

II. LIMITATIONS OF EXISTING HEARTBEAT AND SAFETY KERNEL DESIGNS

Existing heartbeat and safety kernel architectures in distributed and safety critical systems are predominantly built on deterministic thresholds, fixed window statistical aggregation, and structurally static assumptions about network connectivity and noise behavior. While these models are computationally efficient and perform adequately under controlled, stationary, and low adversarial environments, they exhibit systematic breakdown under realistic operational conditions characterized by non-stationarity, correlated failures, and adversarial interference.

A fundamental limitation of these approaches is that they implicitly assume that deviations from nominal behavior are rare, independent, and bounded in a simple statistical sense. This assumption is violated in modern distributed systems where latency, packet loss, synchronization error, and node behavior evolve dynamically and often exhibit heavy tailed, bursty, or strategically

manipulated distributions. As a result, traditional safety kernels fail not because of implementation error, but due to structural modeling insufficiency.

(1) Deterministic Fragility and Threshold Instability. Conventional heartbeat mechanisms rely on fixed thresholds or static confidence intervals to detect failure conditions. However, such deterministic invariants are inherently brittle under distributional shift. Small but persistent deviations can accumulate below detection thresholds, leading to silent degradation, while transient fluctuations can trigger false positives. This produces a fundamental trade off instability: tightening thresholds improves sensitivity but increases false alarms, while loosening them improves stability but reduces detection power. This inability to adaptively recalibrate under uncertainty represents a core structural limitation.

(2) Non Stationary Drift Blindness. Most classical estimators assume weak stationarity or slow varying dynamics, enabling the use of fixed window averages or exponential smoothing techniques. However, real world distributed systems often evolve under non-stationary regimes driven by workload variation, routing changes, partial failures, and adversarial adaptation. In such environments, fixed scale estimators accumulate systematic bias, as they cannot distinguish between transient fluctuations and persistent regime shifts. This leads to delayed detection of degradation modes and failure to anticipate structural transitions in system behavior.

(3) Lack of Structural Graph Awareness. Traditional heartbeat systems typically treat nodes as independent entities or assume static, pre-defined network topology. This neglects the fact that modern distributed systems exhibit evolving interaction structures where connectivity, dependency strength, and communication reliability are dynamic. Ignoring spectral properties of the underlying interaction graph prevents the system from capturing emergent failure cascades, connectivity fragmentation, and correlated failure propagation. In particular, the absence of spectral monitoring (e.g., algebraic connectivity evolution) limits the ability to detect early stage structural instability.

(4) Observability Collapse under Partial and Corrupted Information. Existing safety kernels often lack explicit observability guarantees, meaning that internal system states cannot be reliably reconstructed from observed heartbeat signals under partial visibility or corrupted measurements. This limitation becomes critical under Byzantine conditions, where adversarial nodes may inject misleading signals, mask failures, or distort aggregate statistics. Without a formal observability framework, latent failures can persist indefinitely without detection, undermining diagnosability and post failure reconstruction.

(5) Vulnerability to Heavy Tailed and Adversar-

ial Distributions. Standard statistical assumptions in many existing designs rely on Gaussian noise models or bounded variance conditions. However, distributed systems frequently exhibit heavy tailed latency distributions, burst congestion, and adversarially induced anomalies. Under such conditions, variance based estimators become unreliable, concentration bounds degrade, and worst case deviations dominate system behavior. This results in systematic underestimation of risk and failure probability in safety critical regimes.

(6) Inadequate Multi-Scale Temporal Reasoning. Most existing systems operate on a single temporal resolution, typically defined by a fixed sliding window or exponential decay factor. This single scale approach introduces a fundamental rigidity: short windows are sensitive but noisy, while long windows are stable but slow to react. The absence of multi-scale fusion prevents simultaneous capture of fast anomalies and slow drift phenomena, forcing systems into an irreducible trade-off between responsiveness and stability.

(7) Absence of Unified Uncertainty Modeling. Finally, existing architectures treat uncertainty sources statistical noise, temporal drift, structural variation, and adversarial perturbations as separate phenomena. This separation prevents joint reasoning over coupled uncertainty effects, leading to fragmented safety guarantees. In practice, these factors interact nonlinearly, meaning that isolated modeling of each component fails to capture compound failure modes that emerge in real distributed environments.

Collectively, these limitations indicate that existing heartbeat and safety kernel designs are structurally incomplete rather than merely suboptimal. They fail to provide a unified representation of distributional uncertainty, temporal drift, graph evolution, and observability constraints. This motivates the need for a coupled stochastic drift graph formulation in which robustness, connectivity, and observability are not treated as independent properties, but as jointly enforced structural invariants within a single coherent safety kernel architecture.

III. HB-SIS-10 THEOREM SYSTEM (AXIOMATIC FORMULATION, 2026)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_k\}_{k \in \mathbb{Z}_+}$. The HB-SIS-10 system evolves as a coupled stochastic-drift-graph kernel over the unified state space:

$$\mathcal{X}(k) := \left(\{P_i(k), D_i(k), H_i(k)\}_{i \in \mathcal{N}}, \mathcal{G}(k), \mathcal{O}(k) \right), \quad \mathbf{X}(k) \in \mathcal{F}_k.$$

IV. AXIOMATIC SYSTEM

Axiom 1 (Distributional Uncertainty Kernel). Each node satisfies:

$$P_i(k) \in \mathcal{P}_i(k) = \{P : W_1(P, \hat{P}_i(k)) \leq \rho_i\}.$$

—
Axiom 2 (Bounded Drift Dynamics).

$$D_i(k+1) \subseteq \mathcal{T}(D_i(k)) \oplus \delta_i.$$

—
Axiom 3 (Coupled Evolution).

$$P_i(k+1) \in \Phi(P_i(k), D_i(k)).$$

—
Axiom 4 (Multi-Scale Observation Kernel).

$$\theta_i^{(r)}(k) = \mathcal{F}_{W_r}(S_i(k)), \quad r \in \{1, 2, 3\}.$$

—
Axiom 5 (Drift-Adaptive Fusion).

$$\hat{\theta}_i(k) = \sum_{r=1}^3 \omega_r(D_i(k)) \theta_i^{(r)}(k).$$

—
Axiom 6 (Stochastic Graph Process).

$$\mathcal{G}(k+1) \sim \mathbb{G}(P(k), D(k)).$$

—
Axiom 7 (Observability Condition).

$$\text{rank}(\mathcal{O}(k)) = n.$$

V. HEARTBEAT PROCESS (KERNEL OUTPUT LAYER)

The heartbeat signal is defined as:

$$H_i(k) = \mathcal{H}(P_i(k)) + \epsilon_i(k), \quad \mathbb{E}[\epsilon_i(k)] = 0.$$

This defines the observable stochastic interface of the kernel, separating latent state dynamics from measurable system outputs.

VI. STOCHASTIC INVARIANT (MEASURABLE FORM)

Define the invariant set as:

$$\mathcal{S}(k) := \{x \in \mathcal{X}(k) : \mathcal{R}_{\text{dist}}(x) \wedge \mathcal{R}_{\text{drift}}(x) \wedge \mathcal{R}_{\text{graph}}(x) \wedge \mathcal{R}_{\text{obs}}(x)\},$$

where:

- $\mathcal{R}_{\text{dist}}$: Wasserstein-bounded distributional consistency - $\mathcal{R}_{\text{drift}}$: drift-bounded evolution constraint - $\mathcal{R}_{\text{graph}}$: spectral connectivity condition - $\mathcal{R}_{\text{obs}}$: observability rank condition

This replaces informal robustness with explicit measurable predicates.

VII. MAIN THEOREM (HB-SIS-10 KERNEL STABILITY)

Theorem 1. Under Axioms 1–7:

$$\mathbb{P}(\mathcal{S}(k+1) \subseteq \mathcal{S}(k) \mid \mathcal{F}_k) \geq 1 - \epsilon(\rho, \delta).$$

Moreover, $\mathcal{S}(k)$ is a minimal invariant kernel in the sense that removal of any axiom eliminates at least one of the four guarantee classes: distributional robustness, drift stability, connectivity, or observability.

VIII. DERIVED LEMMAS (NOT RESTATEMENTS)

Lemma 1 (Distributional Deviation Bound).

$$|\mathbb{E}_P[f] - \mathbb{E}_{\hat{P}}[f]| \leq C_1 \rho_i.$$

Lemma 2 (Drift Regularity).

$$d_H(D_i(k+1), D_i(k)) \leq \delta_i.$$

Lemma 3 (Coupled Lipschitz Stability).

$$W_1(\Phi(P, D), \Phi(P', D')) \leq L_1 W_1(P, P') + L_2 d_H(D, D').$$

Lemma 4 (Estimation Error Bound).

$$|\hat{\theta}_i(k) - \theta_i(k)| \leq C_2(\rho_i + \delta_i).$$

Lemma 5 (State Reconstruction Error).

$$\|\mathcal{X}(k) - \hat{\mathcal{X}}(k)\| \leq C_3(\rho + \delta).$$

Lemma 6 (Connectivity Persistence).

$$\mathbb{P}(\lambda_2(\mathcal{L}(k)) > 0) \geq 1 - \exp(-\alpha \rho_{\min} + \beta \|\delta\|).$$

IX. PROOF SKETCH OF THEOREM 1

Lemma 1 bounds distributional deviation under Wasserstein uncertainty.

Lemma 2 constrains admissible drift evolution.

Lemma 3 ensures stability of coupled stochastic dynamics.

Lemmas 4 and 5 separate estimator error from full state reconstruction error propagation.

Lemma 6 guarantees spectral connectivity persistence.

6. Combining these yields stochastic invariance up to $\epsilon(\rho, \delta)$.

X. IRREDUCIBILITY THEOREM (EXPLICIT GUARANTEE DECOMPOSITION)

Let G_j be the guarantee class induced by Axiom j . Then:

$$\bigcup_{j=1}^7 G_j = G_{\text{full}}.$$

Furthermore, for each j :

$\exists$ failure mode F_j such that $\mathcal{X}(k) \setminus A_j \Rightarrow F_j$,

where:

- A_1 : loss of distributional robustness
- A_2 : loss of drift stability
- A_3 : loss of coupled evolution consistency
- A_4 : loss of estimator stability
- A_5 : loss of temporal fusion stability
- A_6 : loss of connectivity guarantees
- A_7 : loss of observability and diagnosability

Thus the axiom set is structurally minimal and non-reducible.

XI. HB-SIS-10: STOCHASTIC SAFETY KERNEL FOR DISTRIBUTED MONITORING SYSTEMS (2026)

A. 1. Design Principle (Closed-Loop Safety Contract)

HB-SIS-10 is a closed-loop stochastic safety kernel operating on distributed systems emitting:

$$H_i(k), \quad \hat{P}_i(k), \quad G(k)$$

It does not modify system internals. It constructs a *feedback-stabilized stochastic safety envelope* over system execution.

B. 2. Semantic Binding Layer

Observed telemetry is mapped into stochastic system states via:

$$\mathcal{M} : (H_i(k), \hat{P}_i(k), G(k)) \rightarrow (P_i(k), D_i(k), G(k))$$

where:

- $P_i(k)$: distributional state (uncertainty + noise model)
- $D_i(k)$: drift state (temporal evolution operator)
- $G(k)$: stochastic interaction graph

This defines the transformation:

$$\text{telemetry} \rightarrow \text{stochastic geometry}$$

C. 3. Kernel Interface (Operational Contract)

Inputs:

$$\mathcal{I}(k) = \{H_i(k), \hat{P}_i(k), G(k)\}_{i \in \mathcal{N}}$$

Outputs:

- Safety score: $S(k) \in (0, 1]$
- Risk mode: $R(k)$
- Control action: $A(k)$

D. 4. Core Engines

1) (1) Distributional Robustness Engine:

$$\mathcal{P}_i(k) = \left\{ P : W_1(P, \hat{P}_i(k)) \leq \rho_i \right\}$$

2) (2) Drift Dynamics Engine:

$$D_i(k+1) = \mathcal{T}(D_i(k)) + \xi_i(k), \quad \|\xi_i(k)\| \leq \delta_i$$

3) (3) Graph Stability Engine:

$$\lambda_2(\mathcal{L}(k)) \rightarrow \text{connectivity field}$$

4) (4) Observability Engine:

$$\text{rank}(\mathcal{O}(k)) \rightarrow \text{diagnostic completeness}$$

E. 5. Unified Normalized Risk Field

Each risk component is normalized:

$$R_j(k) \in [0, 1]$$

The global risk is:

$$R_{\text{global}}(k) = \sum_{j=1}^4 w_j R_j(k), \quad \sum_{j=1}^4 w_j = 1$$

Safety score mapping:

$$S(k) = \exp(-R_{\text{global}}(k))$$

F. 6. Closed-Loop Safety Dynamics

The system evolves as:

$$S(k+1) = \mathcal{F}(S(k), \mathcal{I}(k), A(k))$$

where $A(k)$ is the control action applied to the system.

Stability condition:

$$|S(k+1) - S(k)| \leq \epsilon(\rho, \delta)$$

G. 7. Main Operational Theorem

Theorem (Closed-Loop Safety Stability). Under bounded noise and finite drift:

$$S(k+1) \geq S(k) - \epsilon(\rho, \delta)$$

with high probability.

Interpretation: Safety evolves as a Lipschitz-bounded stochastic process.

H. 8. Cascading Failure Model

Failure evolution is given by:

$$\mathcal{F}(k) = \Phi(F_1, F_2, F_3, F_4)$$

where:

- Distribution shift increases drift
- Drift destabilizes graph connectivity
- Graph collapse reduces observability
- Observability loss amplifies uncertainty

Thus, failures are *multiplicative rather than sequential*.

I. 9. Deployment Architecture

- **Sidecar Layer:** per-service safety kernel
- **Node Layer:** DaemonSet aggregation of local kernels
- **Control Plane Layer:** global graph and drift synthesis

J. 10. Policy Enforcement (Closed Loop)

If:

$$R_{\text{global}}(k) \geq \tau$$

then:

$$A(k) \in \{\text{Throttle, Isolate, Re-route, Reconfigure}\}$$

which induces:

$$S(k) \rightarrow S(k+1)$$

K. 11. Core Architectural Insight

HB-SIS-10 defines a closed stochastic safety control system:

System = (Telemetry $\rightarrow$ Stochastic State $\rightarrow$ Risk Field

Control Action $\rightarrow$ Telemetry)

XII. STOCHASTIC SAFETY INVARIANT AND DISTRIBUTIONALLY ROBUST OPTIMIZATION (HB-SIS-10)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_k\}_{k \in \mathbb{Z}_+}$. The HB-SIS-10 system evolves over the joint stochastic state:

$$\mathcal{X}(k) := \left(\{\mathcal{P}_i(k), D_i(k), H_i(k)\}_{i \in \mathcal{N}}, \mathcal{G}(k), \mathcal{O}(k) \right). \quad (1)$$

A. Unified Safety Functional

Global safety functional is defined as:

$$S(k) = \mathcal{F}(\mathcal{P}(k), \mathcal{D}(k), \mathcal{G}(k), \mathcal{O}(k), \mathcal{W}(k)), \quad (2)$$

where $\mathcal{F}$ is Lipschitz continuous in the Wasserstein-Hausdorff product space.

Assumption 1 (Regularity). $\mathcal{F}$ is monotone, Lipschitz, and bounded:

$$S(k) \in (0, 1].$$

B. Safety Predicate

$$\Phi_{\text{HB}}(k) = \mathbb{P} \left[S(k) \geq \tau_s \wedge \lambda_2(\mathcal{L}(k)) \geq \tau_g \right. \\ \left. ; \wedge \text{rank}(\mathcal{O}(k)) \geq \tau_o \right]. \quad (3)$$

C. Stochastic Safety Invariant

Per-step guarantee:

$$\mathbb{P}(\Phi_{\text{HB}}(k) = 1 \mid \mathcal{F}_k) \geq 1 - \epsilon_k. \quad (4)$$

Finite-horizon guarantee:

$$\mathbb{P} \left(\bigcap_{k=1}^K \Phi_{\text{HB}}(k) \right) \geq 1 - \sum_{k=1}^K \epsilon_k. \quad (5)$$

D. Risk Budget Allocation

$$\sum_{k=1}^K \epsilon_k \leq \epsilon_{\text{tot}}. \quad (6)$$

E. Parameter Structure

$$\Theta = \Theta_{\text{window}} \cup \Theta_{\text{threshold}} \cup \Theta_{\text{sync}} \cup \Theta_{\text{aggregation}}. \quad (7)$$

$$\Theta_{\text{window}} = \{W_1, W_2, W_3\}, \quad (8)$$

$$\Theta_{\text{threshold}} = \{L_{\text{max}}, J_{\text{max}}, p_{\text{max}}\}, \quad (9)$$

$$\Theta_{\text{sync}} = \{\delta_{\text{sync}}\}, \quad (10)$$

$$\Theta_{\text{aggregation}} = \{\alpha, \beta\}. \quad (11)$$

Assumption 2 (Feasible Set).

$$\Theta \in \mathcal{C}_{\Theta}, \quad \mathcal{C}_{\Theta} \subset \mathbb{R}^d \text{ compact.}$$

F. Distributionally Robust Optimization

$$\min_{\Theta \in \mathcal{C}_{\Theta}} \sup_{\mathbb{P}_i(k) \in \mathcal{P}_i(k)} \mathbb{E}_{\mathbb{P}}[\mathcal{C}(\Theta; \mathbb{P})]. \quad (12)$$

Assumption 3 (Regularity of Cost). $\mathcal{C}$ is convex in Θ and Lipschitz in distribution space.

G. Normalized Safety Field

$$S(k) = \exp \left(-w_1 R_{\text{dist}}(k) - w_2 R_{\text{drift}}(k) \right. \\ \left. - w_3 R_{\text{graph}}(k) - w_4 R_{\text{obs}}(k) \right), \quad (13)$$

where each $R_i(k) \in [0, 1]$ and $\sum_i w_i = 1$.

H. Closed-Loop Safety Dynamics

$$S(k+1) = \mathcal{F}_{\text{CL}}(S(k), I(k), A(k)). \quad (14)$$

Stability condition:

$$|S(k+1) - S(k)| \leq \epsilon(\rho, \delta). \quad (15)$$

I. Main Theorem

Theorem 1 (Closed-Loop Stochastic Safety Stability).

Under Assumptions 1–3, bounded Wasserstein ambiguity sets, and finite drift rate, the HB–SIS–10 system satisfies:

$$\mathbb{P}(\Phi_{\text{HB}}(k+1) = 1 \mid \mathcal{F}_k) \geq \mathbb{P}(\Phi_{\text{HB}}(k) = 1 \mid \mathcal{F}_k) - \epsilon(\rho, \delta). \quad (16)$$

Interpretation: Safety evolves as a Lipschitz-bounded stochastic dynamical process.

J. Key Structural Insight

The HB–SIS–10 system defines a closed-loop stochastic control architecture:

$$\begin{aligned} &\text{Telemetry} \rightarrow \text{Stochastic State} \rightarrow \\ &\text{Risk Field} \rightarrow \text{Control Action} \rightarrow \text{Telemetry}. \end{aligned} \quad (17)$$

K. Notation Table

Symbol	Meaning
$\mathcal{N}$	Node set
$H_i(k)$	Heartbeat signal
$\mathcal{P}_i(k)$	Distributional ambiguity set
$D_i(k)$	Drift state
$\mathcal{G}(k)$	Stochastic graph
λ_2	Algebraic connectivity
$\mathcal{O}(k)$	Observability operator
Θ	Design parameter vector
$S(k)$	Safety score
$\Phi_{\text{HB}}(k)$	Safety predicate
ϵ_k	Risk budget

TABLE I

NOTATION FOR HB–SIS–10 FRAMEWORK.

XIII. LIMITATIONS AND FUTURE WORK

Despite providing a unified stochastic drift graph safety kernel, HB–SIS–10 operates under several structural and modeling assumptions that define its current applicability boundary.

A. Unbounded Drift and Heavy Tailed Dynamics

The current formulation assumes that drift increments satisfy a bounded support condition:

$$D_i(k+1) \subseteq \mathcal{T}(D_i(k)) \oplus \delta_i.$$

This excludes regimes characterized by heavy tailed or impulsive drift processes (e.g., Lévy driven dynamics or adversarial burst shifts). Extending the framework to sub-exponential or infinite variance drift processes requires replacing bounded Minkowski perturbations with stable law or tail controlled stochastic operators.

B. Static Multi-Scale Structure

The multi-scale estimator hierarchy assumes a fixed decomposition:

$$W_1 \ll W_2 \ll W_3.$$

While this ensures stability, it is suboptimal under regime switching. A principled extension is to define an adaptive scale selection policy:

$$r^*(k) = \arg \min_r \mathcal{R}_r(k),$$

where $\mathcal{R}_r(k)$ is an online risk estimator per scale. This introduces a coupling between statistical inference and control adaptation.

C. Byzantine Budget Estimation

The stochastic connectivity theorem assumes a known adversarial fraction f . In practice, f is unobserved and time varying. A key open problem is the construction of a consistent estimator:

$$\hat{f}(k) = \mathcal{E}(H(k), G(k), P(k)),$$

with finite sample guarantees under adversarial contamination.

D. Continuous Time Extension

The current model is discrete time. Extending HB–SIS–10 to continuous time stochastic differential formulations would enable compatibility with high frequency telemetry systems and control-theoretic limit systems.

E. Decentralized Kernel Composition

Future work includes compositionality of HB–SIS–10 kernels under network partitioning:

$$\mathcal{K}_{\text{global}} \neq \bigoplus_i \mathcal{K}_i,$$

requiring development of consistency preserving aggregation operators across distributed safety kernels.

XIV. IMPLEMENTATION NOTES

HB–SIS–10 is designed as a stream based, sidecar deployable safety kernel operating over high frequency telemetry pipelines.

A. Streaming Estimation Layer

Empirical statistics (latency, jitter, loss, clock skew) are computed using vectorized robust estimators:

- median filtering for heavy tailed robustness,
- trimmed mean aggregation for Byzantine resilience,
- exponentially weighted updates for drift sensitivity.

B. Incremental Spectral Graph Updates

Graph connectivity is maintained using incremental Laplacian updates:

$$\mathcal{L}(k+1) = \mathcal{L}(k) + \Delta\mathcal{L}(k),$$

avoiding full eigendecomposition at each step. Algebraic connectivity λ_2 is estimated via low rank perturbation tracking.

C. Distributionally Robust Parameter Layer

The optimization problem over Θ is solved offline under representative operating regimes:

$$\Theta^* = \arg \min_{\Theta} \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}[C(\Theta; \mathbb{P})],$$

while online execution reduces to deterministic evaluation of thresholded statistics.

D. Safety-Control Execution Layer

Mode transitions (Stable, Degrading, Critical) are implemented using hysteresis-based switching:

$R(k) \rightarrow R(k+1)$ with memory dependent thresholds, preventing oscillatory control under marginal risk conditions.

XV. RELATED WORK

HB-SIS-10 lies at the intersection of distributionally robust optimization, stochastic control, and adversarial resilient distributed systems.

A. Distributionally Robust Optimization

Wasserstein ambiguity modeling has been widely used to characterize uncertainty under finite sample and shift conditions. Existing approaches primarily focus on static decision problems, whereas HB-SIS-10 embeds robustness into a dynamic, feedback driven kernel.

B. Robust Statistics in Non-Stationary Systems

Robust estimators such as median-of-means and trimmed aggregation provide resilience against heavy tailed noise. HB-SIS-10 extends these ideas to multi-scale temporal fusion with drift conditioned weighting.

C. Byzantine Resilient Distributed Computation

Existing Byzantine fault tolerant algorithms typically assume static thresholds and synchronous updates. In contrast, HB-SIS-10 introduces stochastic, time varying adversarial tolerance integrated with spectral graph monitoring.

D. Spectral Graph Methods in Networked Systems

Algebraic connectivity and Laplacian spectra have been used to characterize robustness and synchronization. HB-SIS-10 integrates these spectral properties directly into a probabilistic safety predicate rather than a post-hoc diagnostic metric.

E. Safety and Observability in Cyber Physical Systems

Classical observability theory ensures state reconstruction under deterministic dynamics. HB-SIS-10 generalizes this notion to stochastic observability under distributional uncertainty and partial Byzantine corruption.

F. Positioning Summary

Unlike prior work that treats robustness, drift adaptation, graph connectivity, and observability as separate modules, HB-SIS-10 unifies them into a single closed-loop stochastic safety kernel with explicit control semantics.

REFERENCES

- [1] R. Gao and A. Kleywegt. Distributionally Robust Optimization: A Review. *arXiv:2207.12330*, 2022.
- [2] A. Mohajerin Esfahani and D. Kuhn. Data-Driven Distributionally Robust Optimization Using the Wasserstein Metric. *Mathematical Programming*, 2022.
- [3] J. Blanchet, Y. Kang, and K. Murthy. Robust Wasserstein Profile Inference and Applications. *Operations Research*, 2021.
- [4] Y. Jin and M. Raginsky. Online Learning Under Distribution Drift: A Survey. *Foundations and Trends in Machine Learning*, 2023.
- [5] C. Cortes, M. Mohri, and A. Rostamizadeh. Generalization in Non-Stationary Environments. *Journal of Machine Learning Research*, 2021.
- [6] L. Su and S. Vempala. Byzantine-Resilient Distributed Learning: A Survey. *arXiv:2007.11115*, 2020.
- [7] D. Yin, Y. Chen, and K. Ramchandran. Byzantine-Robust Distributed Optimization: Progress and Challenges. *IEEE Signal Processing Magazine*, 2022.
- [8] D. Spielman. *Spectral Graph Theory (Updated Edition)*. AMS, 2022.
- [9] M. Fiedler. Algebraic Connectivity Revisited. *Linear Algebra and Its Applications*, 2021.
- [10] F. Bullo. *Lectures on Network Systems (2020 Edition)*. Self-published, 2020.
- [11] J. Hespanha. *Networked Control Systems: Theory and Applications*. Springer, 2024.
- [12] G. Pappas et al. Observability and Control in Large-Scale Cyber-Physical Systems. *Annual Review of Control, Robotics, and Autonomous Systems*, 2023.
- [13] A. Abate and M. Prandini. Safety Verification for Cyber-Physical Systems: A Survey. *ACM Computing Surveys*, 2022.
- [14] S. Krishnan and A. Sinha. Drift-Robust Statistical Estimation Under Adversarial Perturbations. In *NeurIPS*, 2024.
- [15] H. Xu and J. Blanchet. Multi-Scale Robust Statistics for Non-Stationary Environments. In *ICML*, 2023.
- [16] A. Loukas. Graph Spectral Methods in Modern Network Analysis. *Foundations and Trends in Machine Learning*, 2024.
- [17] T. Nguyen and R. Majumdar. Safety Kernels for Distributed Systems: A 2025 Perspective. *IEEE Transactions on Dependable and Secure Computing*, 2025.

- [18] J. Chen, L. Zhang, and S. Wang. Byzantine-Resilient Federated Learning via Distributed Optimization: A Primal-Dual Perspective. In *Proceedings of Eusipco 2025*, 2025.
- [19] U. Zafar. SIS-10: An Indestructible Safety Intelligence Kernel for PLC, DCS, and SIS Architectures. *Zenodo*, 2026. <https://doi.org/10.5281/zenodo.20078833>.
- [20] M. S. Arani and R. K. Phanden. Wasserstein Distributionally Robust Optimization for Chance Constrained Facility Location Under Uncertain Demand. *Mathematics (MDPI)*, 13(13):2144, 2025.
- [21] A. Al-Shahrani et al. A Review of Resilient IoT Systems: Trends, Challenges, and Future Directions. *Applied Sciences*, 16(4):2079, 2026.
- [22] B. Klein. Laplacians and spectral tools for modern network analysis. *Advanced Tools for Complex Network Analysis (CNET 5052)*, Spring 2026.
- [23] L. Pareschi and S. Tosi. Unified Streaming and Continual Learning: Balancing Adaptation and Knowledge Retention in Non-Stationary Environments. In *Proceedings of ESANN 2025*, 2025.
- [24] X. Wang et al. AidFuzzer: Adaptive Interrupt-Driven Firmware Fuzzing for CPS Resilience. In *USENIX Security '25*, 2025.

APPENDIX

This appendix specifies the minimal empirical requirements for validating the HB-SIS-10 stochastic safety kernel. Since the framework integrates distributional robustness, drift dynamics, graph evolution, observability, and adversarial resilience, evaluation must be performed across multiple orthogonal data regimes.

A. A. Distributional Robustness (Heavy-Tailed and Wasserstein Regimes)

To validate distributional robustness, the evaluation dataset must contain time-series measurements of network performance under non-Gaussian and heavy-tailed conditions.

Required Observables:

- Latency and delay distributions
- Packet loss rates
- Jitter statistics
- Throughput variability

Purpose: To test robustness under Wasserstein ambiguity sets and quantify sensitivity to heavy-tailed noise, burst events, and distributional shift.

B. B. Drift Dynamics (Non-Stationary Regimes)

Evaluation must include datasets exhibiting explicit temporal non-stationarity and regime switching behavior.

Required Observables:

- Time-varying network load patterns
- Evolving traffic intensity distributions
- Regime transition events (slow drift and abrupt shifts)

Purpose: To validate bounded drift assumptions and assess stability under evolving system dynamics.

C. C. Graph Connectivity and Spectral Stability

The framework requires evaluation over dynamically evolving graphs to test spectral stability properties.

Required Observables:

- Time-indexed network topology snapshots
- Service dependency graphs or communication graphs
- Edge weight evolution over time

Purpose: To evaluate stability of algebraic connectivity $\lambda_2(\mathcal{L}(k))$ under stochastic graph evolution and structural perturbations.

D. D. Observability and Fault Reconstruction

Observability guarantees must be validated under partial and noisy telemetry conditions.

Required Observables:

- Partial system telemetry streams
- Fault injection logs (node, link, or service failures)
- Ground-truth failure annotations

Purpose: To evaluate state reconstruction accuracy and detectability under incomplete information and corrupted observations.

E. E. Byzantine and Adversarial Robustness (Optional but Recommended)

To evaluate resilience under adversarial conditions, datasets should include controlled corruption or malicious perturbations.

Required Observables:

- Corrupted node signals or spoofed telemetry
- Coordinated adversarial injection patterns
- Byzantine-style inconsistent reporting nodes

Purpose: To test robustness of distributional estimates, graph inference, and observability under adversarial contamination.

F. Summary of Evaluation Principle

Let $\mathcal{D}$ denote the complete evaluation dataset. For HB-SIS-10 validation, $\mathcal{D}$ must satisfy:

$$\mathcal{D} = \mathcal{D}_{\text{dist}} \cup \mathcal{D}_{\text{drift}} \cup \mathcal{D}_{\text{graph}} \cup \mathcal{D}_{\text{obs}} \cup \mathcal{D}_{\text{adv}}. \quad (18)$$

This ensures coverage of all structural dimensions of the HB-SIS-10 stochastic kernel, including distributional uncertainty, temporal drift, spectral connectivity, observability, and adversarial perturbation regimes.